Module-1

Introduction to Artificial Intelligence

Introduction to Artificial intelligence: What is AI, History of AI, Types of AI, Components of Intelligence, Agent and Environment, Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.

1. What is Artificial Intelligence?

Artificial Intelligence (AI), simply means the intelligence demonstrated by machines that help them to mimic the actions of humans. AI simulates natural intelligence in machines that are programmed to learn from experiences, adjust to new inputs and perform human-like tasks.

1.1 Advantages and Disadvantages of Artificial Intelligence

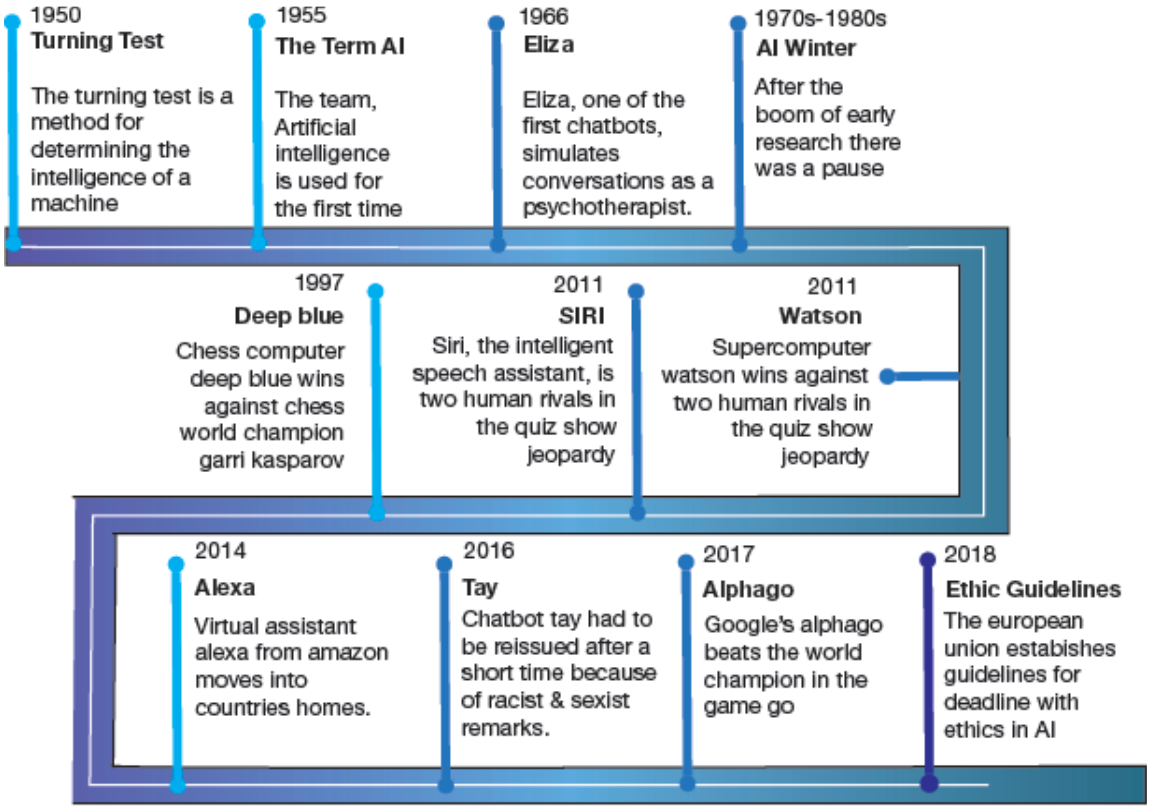
Advantages:

1. Performs well on tasks that uses detailed data.
2. Takes less time to perform tasks that needs to process huge volumes of data.
3. Generates consistent and accurate results.
4. Can be used 24 X 7.
5. Optimizes tasks by better utilizing resources.

Disadvantages:

1. Involves more cost.
2. Technical expertise required to develop and use AI applications.
3. Lack of trained professionals.
4. Incomplete or inaccurate data may result in disastrous results.
5. Lacks the capability to generalize tasks.

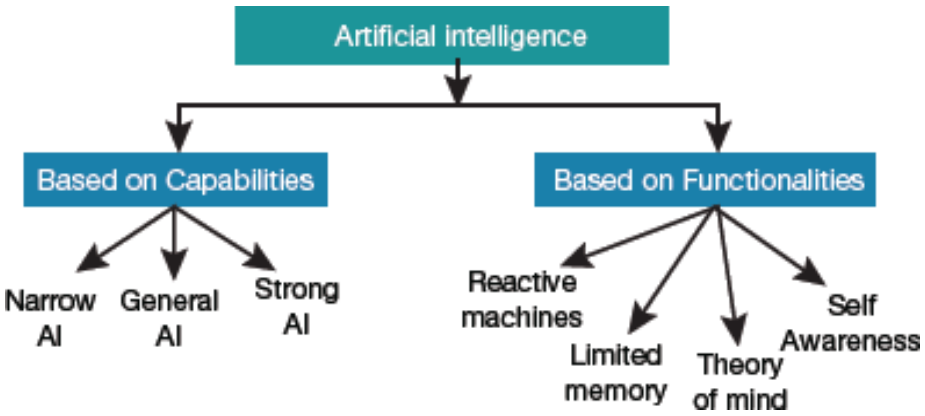
2. History of Artificial Intelligence



- In **1952**, Arthur Samuel developed a self-learning program to play checkers.
- In **1954**, the IBM computer could translate 60 carefully selected Russian sentences into English.
- In **1956**, John McCarthy coined the term 'artificial intelligence'. The scope and goals of AI were discussed in a conference later.
- In **1997**, IBM's Deep Blue machine could beat world chess champion, Gary Kasparov.
- In **2005**, STANLEY, a self-driving car, won the DARPA Grand Challenge. Even the US military started investing in autonomous robots like Boston Dynamics' 'Big Dog' and iRobot's 'PackBot.'

- In **2008**, Google did a remarkable work in speech recognition and introduced the feature in its iPhone app.
- In **2011**, Apple released Siri, an AI-powered virtual assistant that could beused through its iOS operating system.
- In **2012**, Andrew Ng, founder of the Google Brain Deep Learning project, fed 10 million YouTube videos as a training set to a neural network and used deep learning algorithms to enable the neural network to recognize a cat without being told what a cat is.
- In **1958**, John McCarthy developed the AI programming language, Lisp. He also published a paper discussing a complete AI system that could learn from experience as effectively as humans do.
- In **1959**, Allen Newell, Herbert Simon and J.C. Shaw developed the General Problem Solver (GPS), a program designed to imitate human problem solving.
- In **2016** Google DeepMind's AlphaGo defeated world champion Go player Lee Sedol. Go is an ancient Chinese game and winning this game was a major breakthrough in the field of AI.
- In the same year, the first 'robot citizen', a humanoid robot, Sophia was created that could perform facial recognition, verbal communication and facial expression.
- In **2018**, Google released NLP engine BERT that reduced barriers in translation and understanding by machine learning applications. In the same year, Waymo One service was launched that allowed users to request a pick-up from one of the company.
- In **2020**, Baidu released its Linear Fold AI algorithm to develop a vaccine during the early stages of the SARS-CoV-2 pandemic. The algorithm could predict the RNA sequence of t.

3. Types of Artificial Intelligence



1. **Weak AI**:

Weak AI, also known as narrow AI,

is specifically designed to perform a specific type of task. For example, Siri and Alexa are weak AI systems. These systems are already trained with appropriate responses to classify things accordingly. When you instruct Alexa to play a song, it responds by playing that song.

In fact, majority of AI applications that we use today (predicting weather, stock prices, optimizing business, etc.) come under this category. Weak AI systems operate within a limited context and are the most successful realization of artificial intelligence to date. Application of narrow AI has resulted in significant societal benefits. Google search, Image recognition software, self-driving cars and IBM's Watson are some examples of such systems.

2. **Strong AI**:

Strong AI, also known as artificial general intelligence or artificial super intelligence (ASI) or superintelligence, makes full attempt to resemble the human brain. It utilizes cognitive skills and fuzzy logic to perform tasks for which it had not been trained earlier. Such a system needs capability for visual perception, speech recognition, decision-making and translations between languages.

As of now, we see general AI only in sci-fi movies, where machines emulate human intelligence and think strategically to perform complex tasks.Many people fear that intelligent robots would overrun humanity, but experts says that we need not worry about it as it is not likely to happen in the near future.

3. Reactive Machines:

Reactive machines are very basic machines with no memory to store and thus use past experiences to determine future actions. They just perceive the world and react to it.

IBM's Deep Blue, which defeated chess grandmaster Kasporov, is such an example. The AI system sees the pieces on a chess board and responds without referring to any of its prior experiences.

Since it lacks capabilities to store past experiences, a reactive machine cannot improve with practice.

Google's AlphaGo is another example of a game-playing reactive machine. The application evaluates future moves using neural network. This gives it an edge over Deep Blue game.

4. Limited Memory:

Limited memory machines retain data but only for a short period of time and can use it only for a limited time. They cannot keep adding data permanently to a library of their experiences. Such systems are usually used in autonomous vehicles since they need not store data (like *recent speed of nearby cars, distance between cars, the speed limit, etc.)* that can help them navigate roads.

AlphaGo also bested world-class competitors of the game, defeating champion Go player Lee Sedol in 2016.

No doubt, limited memory AI is more complex and is used for better possibilities than reactive machines. These machines continuously train a model to analyse and utilize new data. The model also improves using feedback received from humans or the environment and stored as data.

5. Theory of Mind:

The theory of mind focuses on imitating the human brain. For this, it form representations about the world, starting from thoughts, emotions and memories. However, such systems are only theoretical as of now but they may become reality very soon.

These machines must understand thoughts and emotions that affect the behaviour of one's self. Theory of Mind machines make decisions keeping in view the feelings experienced through self-reflection and determination.

6. Self-Awareness:

Once Theory of Mind is established, then the next step would be incorporating self-awareness in AI systems. Self-awareness in machines enable them to possess human-level consciousness, understand the current state—its own existence in the world and use that information to deduce others feelings.

Self-AI systems will be able to understand what others may need. They could interpret user's feelings by learning not only what they communicate to them but also how they communicate it.

4. Components of Intelligence

- Reasoning
- Learning
- Problem solving
- Perception
- Linguistic intelligence

1. Reasoning

It is the set of processes that is used in making decisions and prediction.There are two types of reasoning—inductive and deductive.

Inductive Reasoning	Deductive Reasoning
It uses specific observations to make broad general statements.	It starts with a general statement to obtain a specific, logical conclusion.
Even if all observations are true, there are still chances that the conclusion can be false.	If something is true for a particular class of things in general, it is also true for all members of that class.
Example: 'Manvi is a topper. Manvi is studious. Therefore, all toppers are studious.'	Example: 'Students who scored at least 80% are eligible for placement. Manan scored 90%. Therefore, Manan is an eligible candidate.'

2. **Learning**

It is a process of gaining knowledge or skill by studying, practicing, being taught or experiencing something. Learning is an ability that is possessed by humans, animals and AI-enabled systems to improve the awareness of the subjects of the study.

Learning can be categorized as follows.

Auditory learning : promotes learning by listening and hearing. For example, students hear recorded lectures to understand a particular concept.

Episodic learning: promotes learning by remembering the sequences of events that one has witnessed or experienced. This is linear and orderly.

Motor learning: helps to learn by precise movement of muscles. For example, picking objects.

Observational learning: triggers learning by watching and imitating others. For example, children often learn by mimicking their parents.

Perceptual learning: is done by recognizing stimuli that one has seen before. For example, identifying and classifying objects and situations.

Relational learning: involves learning to differentiate among various stimuli on the basis of relational properties, rather than absolute properties.

3. **Problem Solving**

It is the process in which one tries to find the desired solution in the present

situation by following the path which is blocked by known or unknown hurdles. Problem solving uses **decision-making to** select the best suitable alternative to reach the desired goal.

4. **Perception**

It is the process of acquiring, interpreting, selecting and organizing sensory information. While humans use sensory organs to perceive their environment, AI systems use data acquired by the sensors to do the same.

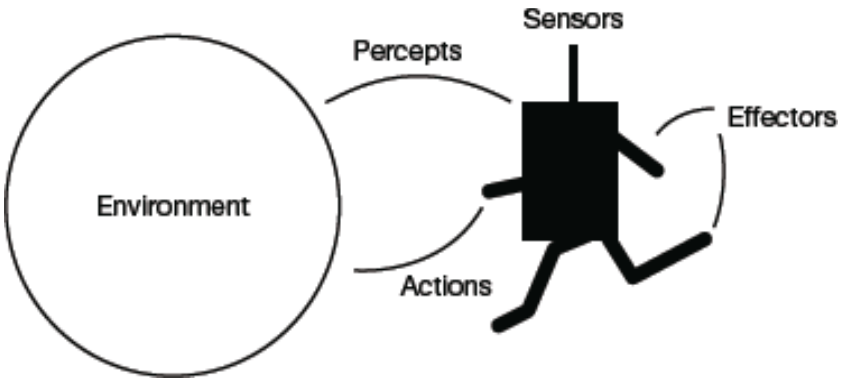
5. **Linguistic Intelligence**

It is used in in interpersonal communication and defines one's ability to use, comprehend, speak and write the verbal and written language.

5. Agent and Environment

Artificial intelligence is all about studying rational agents, which may include anything that makes decisions. For example, a person, firm machine or software that performs action(s) with the best outcome after considering past and current percepts at a given instance are examples of agents. These agents work in their

environment and may interact with other agents in that environment. Thus, we can say that an AI system comprises an **agent and its environment**.



In an artificially intelligent system, the agents act in their environment which may contain other agents.
This **agent** can be anything that is capable of perceiving its environment. For this, it may have **sensors.**

Agents also have effectors that help them to act upon their environment**.**

The different types of agents in an AI system include the following:

1. H**uman agent:** A human agent has sensory organs (like eyes, ears, nose, tongue and skin) that acts as sensors, and other organs such as hands, legs, mouth as effectors for taking actions.

2. R**obotic agent:** A robotic agent uses cameras and infrared range finders as sensors, and various motors and actuators as effectors.

3. **Software agent:** Such an agent uses bit strings as its programs and actions.

6. Types of Agents

1.Simple Reflex Agents

Simple reflex agents choose actions based only on the current percept. They are rational only if they make a correct decision based on the current precept. Their environment is completely observable.

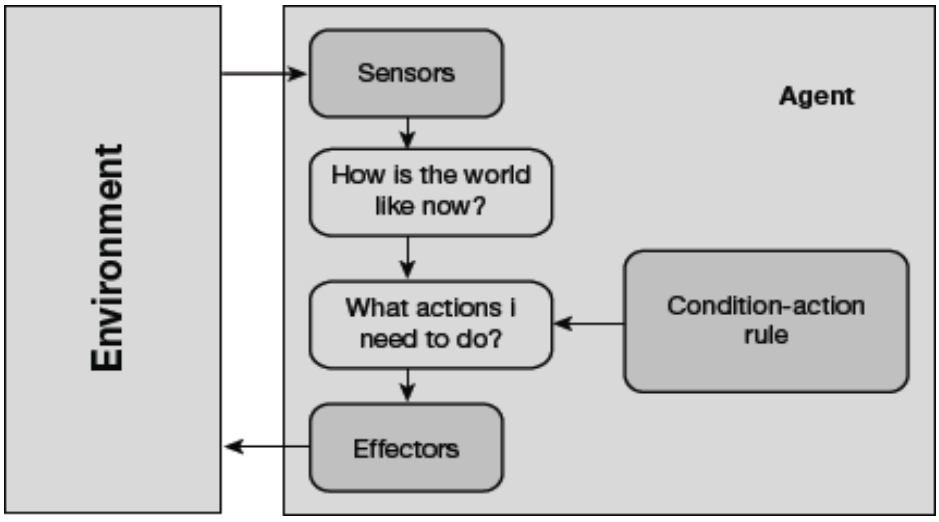
They work using **condition-action rule** that maps a state (condition) to an action. If the condition is true, then the action is taken, else not. However,

the problem with a simple reflex agent is that they succeed only when the environment is fully observable.

1. Possess very limited intelligence
2. Has knowledge of any state other than the current state.
3. Any change in the environment will eventually end up updating the collection of rules.

2.Model-Based Reflex Agents

Model reflex agents use a model of the world to choose their actions. For this, they need to maintain an internal state. Here, **model** means knowledge about 'how things happen in the world'.



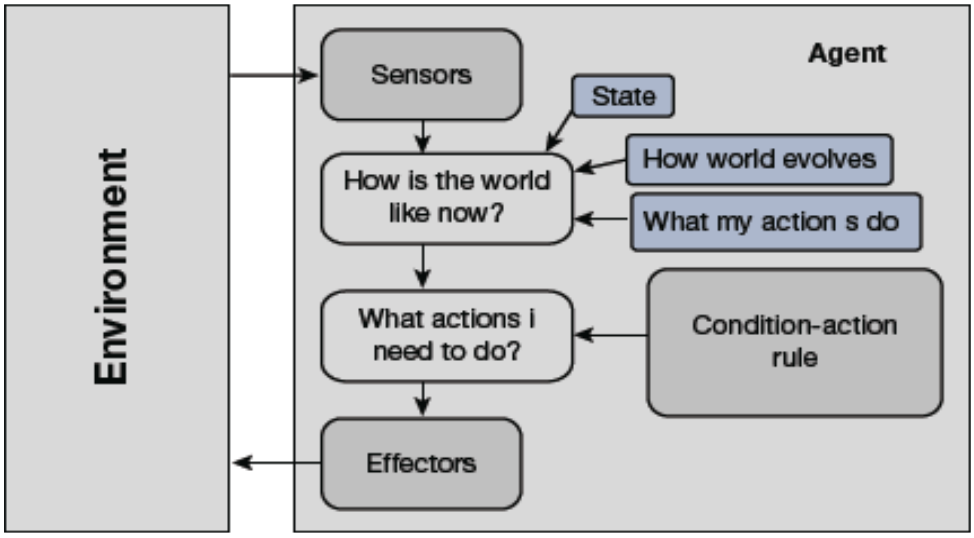
Internal state represents unobserved aspects of the current state depending on percept history.

To update the state, information is required to understand,

- How the world evolves and
- How the agent's actions affect the world.

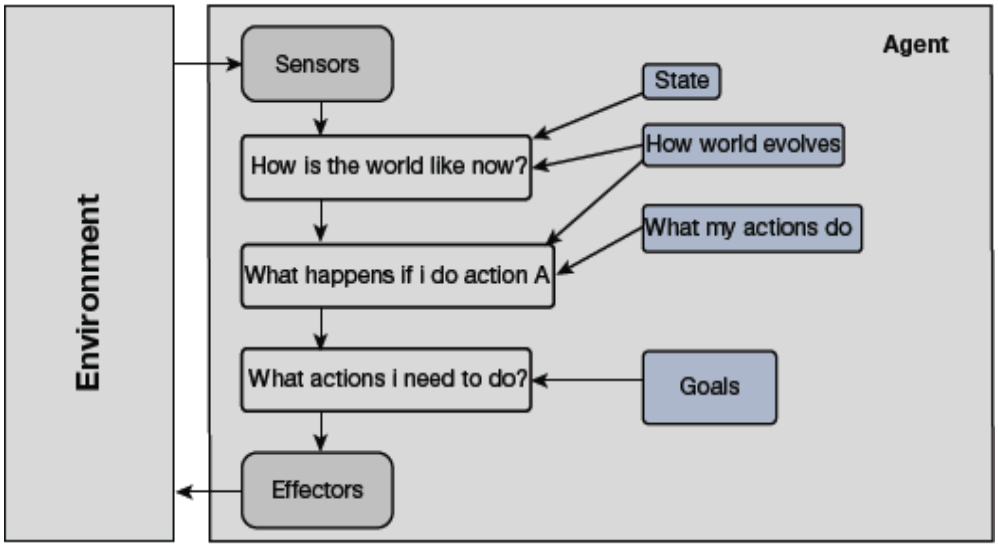
3.Goal-Based Agents

Goal means a description of desirable situations. Goal-based agents choose their actions to achieve goals. This provides more flexibility than reflex agent as the knowledge supporting a decision isexplicitly modeled and permits any sort of modifications.



4.Utility-Based Agents

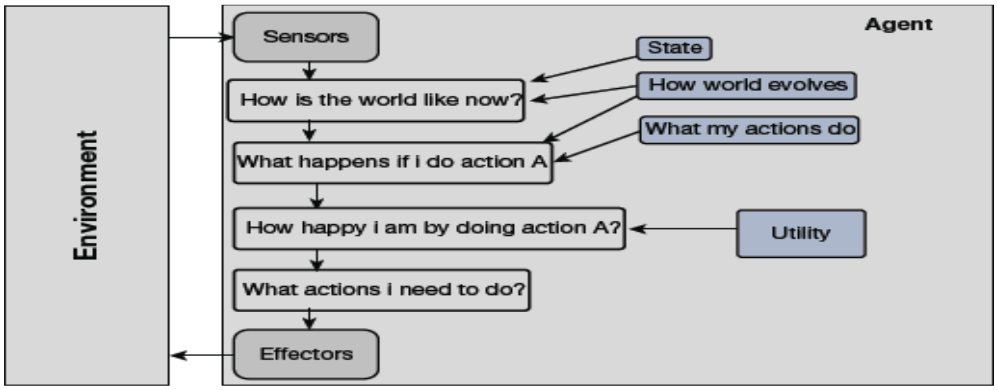
Many a time, goals are either conflicting or are difficult to achieve. In suchsituations, utility-based agents can be used to achieve goals that are more important than others. These agents choose actions based on a preference (utility) for each state.



However, at times, achieving the desired goal is not enough and happiness of the agent is also important. In such a scenario, utility describes how**'happy'** the agent is. A utility agent chooses the action that maximizes the expected utility, that is, the associated degree of happiness.

5.Learning Agent

A learning agent learns from its past experiences. It is designed to possess learning capabilities. Initially, it starts working with basic knowledge and then acts and adapts automatically through learning.



Such an agent has four main conceptual components, as follows:

Learning element, which is responsible for making improvements by learning from the environment

Critic is **t**he element that provides feedback to the learning element, describing how well the agent is doing with respect to a fixed performance standard.

Performance element chooses a particular external action to be taken.

Problem generator suggests actions that result in new and informative experiences.

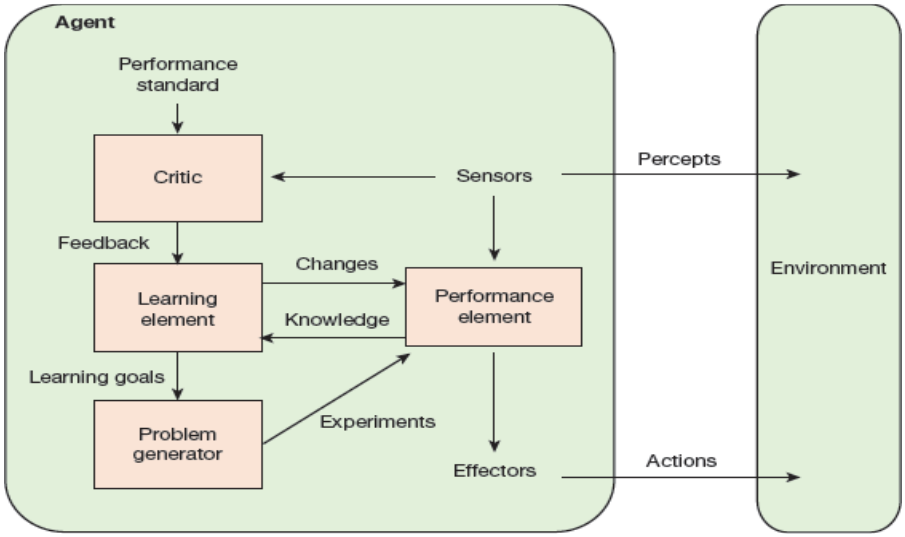


Fig: Citric

7. Types of Environments

As stated earlier, environment is the place where the agent would work. It rewards and specifies states and actions of the agents. Although environment in an AI system is quite vast, we can identify a small number of dimensions to categorize them as given below.

A. Discrete/Continuous

If an environment has a limited number of distinct, clearly defined, states, then it is said to be discrete. For example, a software agent designed to play chess game works in a discrete environment as there are limited moves that a player can make. However, a self-driving car operates in a continuous environment.

While a discrete environment has a finite number of percepts and actions that can be performed within it, a continuous environment has no such constraints.

B. Known Vs Unknown

Known and unknown are based on an agent's state of knowledge to perform an action and they are not actually a feature of an environment. In a known environment, the agent knows the results for all actions but in an unknown environment, the agent has to learn how it should work to perform an action.

Usually, a known environment is partially observable and an unknown environment is fully observable.

C. Observable/Partially Observable

If an agent can determine the complete state of the environment at each time point from the percepts, then it is said to be observable; otherwise, it is only partially observable. Correspondingly, in an unobservable environment, the agent has no sensors. In a fully observable environment, the complete state of the environment *relevant to the choice of action* of the agent can be captured using its sensors.

D. Static/Dynamic

A static environment does not change when an agent is acting, but a dynamic environment does. Agents in static environments, therefore, do not take into account the state of the environment during the action and are thus easy to deal with. However, in a dynamic environment, agents must consider the world while

performing each action. For example, self-driving cars work in a dynamic environment, but games like crossword puzzles and chess operate in a static one.

Between these two extremes, we can also have **semi-dynamic environments** which do not change with the passage of time but the agent's performance score does.

E. Single Agent/Multiple Agents

The environment may contain other agents which may or may not be of the same kind of the agent. For example, vacuum cleaning environment is a single agent but a chess game is a two-agent environment.

In a multi-agent environment, the performance measure of one agent depends on the action performed by another agent. A multi-agent environment may be competitive or cooperative.

F. Accessible/Inaccessible

In an accessible environment, the sensory apparatus of the agent has access to the complete state of the environment. This means that it can obtain complete and accurate information about the state's environment, but this is not so in case of an inaccessible environment.

For example, an empty room whose state can be defined by its temperature is an example of an accessible environment, whereas information about an event on the Earth is an inaccessible environment.

G. Deterministic/Non-deterministic

In a deterministic environment the next state of the environment can be easily determined by the current state as well as the actions of the agent. This is not true for a non-deterministic environment. A n**on-deterministic** environment description maximizes an agent's performance for all possible outcome of its actions. Note that in a deterministic, fully observable environment, the agent does worry about uncertainty.

In a **Stochastic** environment, uncertainty about outcomes is quantified in terms of **probabilities**.

H. Episodic/Non-episodic

In an episodic environment, each episode has an agent perceiving and then acting. The quality of its action depends just on the episode itself. Subsequent episodes do not depend on the actions performed in the previous episodes.

I. Search

Algorithms

We have read that artificial Intelligence is the study of building rational agents. These agents make use of some or the other search algorithms in the background to achieve their tasks. For example, some single-player games like tile games, Sudoku, crossword, etc., use search algorithms to deduce a particular position in the game.

A search problem consists of the following:

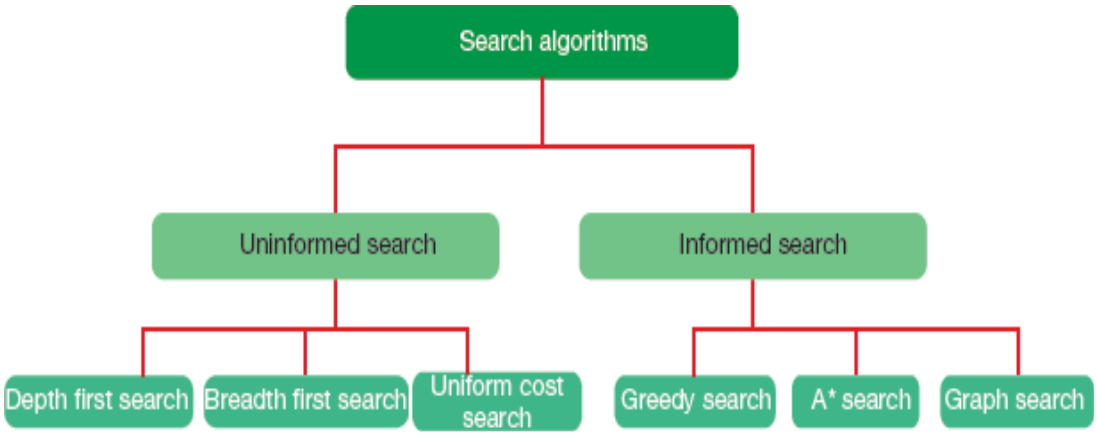
1. **State space:** is the set of all possible states that can be attained by an agent.

2. **Start state:** is the state from where the searching is done.

3. **Goal test:** is a function that checks if the current state is the goal state or not.

4. **Solution:** to a search problem is a sequence of actions (also known as **plan)** that transforms the start state to the goal state. This plan is realized using search algorithms.

7.1 Types of Search Algorithms



1. Informed Search Algorithms:

The informed search algorithm which is extensively used in a large space is also known as heuristic search as it uses a heuristic function. The heuristic function takes the current state of the agent as its input and estimates how close the agent is from the goal. Though the heuristic method may not always give the best solution, it is, however, guaranteed to find a good solution in reasonable time.

A heuristic function h(n), calculates the cost of an optimal path between the pair of states. The value of the heuristic function is always positive. We can write,

h(n) <= h* (n)

where h(n) is heuristic cost, and h (n) is the estimated cost. This means that the heuristic cost should be less than or equal to the estimated cost**.**

7.1.1 Pure Heuristic Search

To solve large problems with a large number of possible states, problem specific knowledge needs to be added to increase the efficiency of search algorithms. In such a situation, heuristic search algorithm performs well.

Heuristic search is the simplest form of heuristic search algorithms that expands nodes based on their heuristic value h(n). The algorithm maintains two lists—OPEN and CLOSED. All nodes that have already been expanded are placed in the CLOSED list, while others that have not been expanded are in the OPEN list.

a. Best-First Search Algorithm (Greedy Search)

Greedy best-first search algorithm combines depth-first search and breadth first search algorithms and always selects the path which appears best at that moment. It uses the heuristic function and best-first search at each step to choose the most promising node. The most promising node is the one that

is closest to the goal node. The greedy best first algorithm is implemented by the priority queue.

Best first search algorithm steps:

Step 1: Insert the starting node in the OPEN list.

Step 2: Stop and return failure if the OPEN list is empty,

Step 3: From the OPEN list, remove the node having the lowest value of h(n), and insert it in the CLOSED list.

Step 4: Expand the node (removed in Step 3) and generate its successors.

Step 5: Check each successor to find if it is the goal node or not. If any successor node is found to be the goal node, then return success and terminate the search, else go to Step 6.

Step 6: Check if each successor node is in either OPEN or CLOSED list. If the node is not present in any of the lists, then add it to the OPEN list.

Step 7: Go to Step 2.

Advantages

1. It takes advantage of both BFS and DFS algorithms.

2. Best first search algorithm is more efficient than BFS and DFS algorithms.

Disadvantages

1. In some worst-case scenarios, best first search can behave as an unguided depth-first search.

2. Like DFS, best first search algorithm can get stuck in a loop.

3. Best first search algorithm is not an optimal algorithm.

Example: Consider the tree given below and traverse it using greedy best first search algorithm.

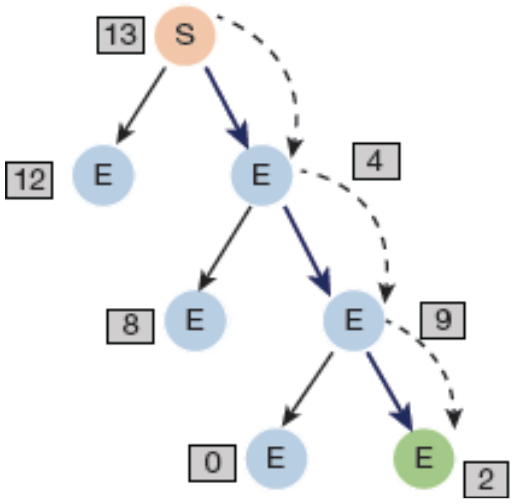


Table shows the heuristic estimates:

Node	h(n)	Node	h(n)	Node	h(n)
A	11	E	4	I, J	3
B	5	F	2	S	15
C, D	9	H	7	G	0

Solution: Expand the node S and put it in the CLOSED list. Generate its successors and place them in the OPEN list.

Open [A, B], Closed [S]

Iteration 1: Since h(n) for node B is less than that of node A, expand B and place it in CLOSED list. Now only node A is in the OPEN list. Open [A], Closed [S, B]

Iteration 2: Generate successors of node B and place them in the OPEN list. Open [E, F, A], Closed [S, B].

Since node F has the lowest heuristic value among all nodes in the OPEN list, we will expand node F. Now, Open [E, A], Closed [S, B, F].

Iteration 3: Generate successors of node F and place them in the OPEN list. So, Open [I, G, E, A], Closed [S, B, F].

Now, generate succors of node F and place them in OPEN list. Since one of the successors of node F is the goal node, the algorithm returns success and terminates. The path is given by the nodes present in the CLOSED list.

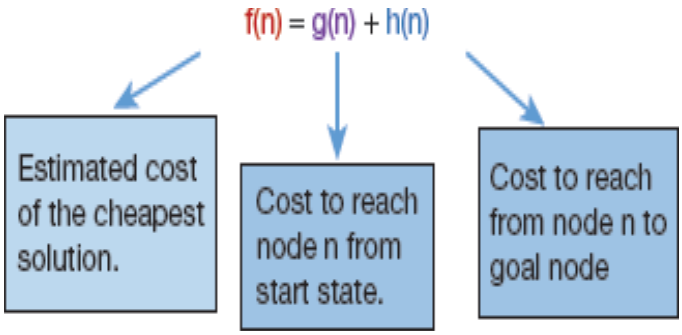
Open [I, E, A], Closed [S, B, F, G]

Hence, the final solution path will be: **S----> B----->F----> G**

The worst-case time complexity and space complexity of greedy best first search is O(b**m**) where, m is the maximum depth of the search space.

b. A* Search Algorithm

The A search algorithm, best-known form of best first search, uses heuristic function h(n), and g(n) which specifies the cost to reach the node n from the start state. It solves the searching problem efficiently by expanding less search tree to find shortest path and give an optimal result faster. That is, it avoids expanding paths that are already expensive, but expands most promising paths first.



The A search algorithm, uses search heuristic as well as the cost to reach the node. Sum of both the costs gives a number known as a **fitness number**.

Step1: Put the starting node in the OPEN list.

Step 2: If the OPEN list is empty then return failure and stop.

Step 3: From the OPEN list, choose the node having minimum value of f(n). If this node is the goal node, then return success and stop.

Step 4: Expand the node selected in Step 3 and generate its successors. Put the node in the closed list. For each successor, check if it is present in the OPEN or CLOSED list. If it is not present in either then compute its f(n)and place it into Open list.

Step 5: Else if the successor node is already in OPEN or CLOSED list, then attach it to the back pointer reflecting the lowest g(n') value.

Step 6: Return to **Step 2**.

Advantages

1. It is better than all other search algorithms.

2. A search algorithm is optimal and complete.

3. The algorithm can be used to solve very complex problems.

4. It is used by many games and web-based maps to find the shortest path efficiently.

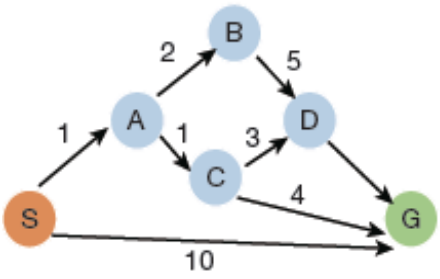
Disadvantages

1. It may not always produce the shortest path since it is based on heuristics and approximation.

2. A search algorithm is more complex than other search algorithms

3. It uses more memory as it keeps all generated nodes in the memory. This makes it difficult to use for large-scale problems.

Example: Consider the graph given below and traverse it using the A algorithm. Note than h(n) will be calculated by adding h(n) and the cost of reaching that node.



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

Solution:

Initially, we start with node S having h(n) = 5. Therefore, {(S, 5)}

Iteration1: From node S, we can reach node A and node G.

h(n) of node A is 3 and the cost to reach node A from node S is 1. Therefore, f(n) of node A = 4. Similarly, node G has h(n) = 0 and the cost to reach node

G from node S is 10. Therefore, f(n) of node G = 10.

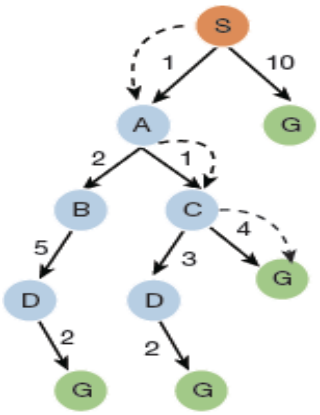
{(S--> A, 4), (S-->G, 10)}

Iteration2: Since node A has less f(n) as compared to node G, we will expand node A.

h(n) of node B is 4 and the cost to reach node B from node S is 3.

Therefore, f(n) of node B = 7. Similarly, node C has h(n) = 2 and the cost to reach node C from node S is 2. Therefore, f(n) of node C = 2.

{(S--> A-->C, 4), (S--> A-->B, 7), (S-->G, 10)}



Iteration3: Since node C has less f(n) as compared to other nodes, we will expand node C.

h(n) of node D is 6 and the cost to reach node D from node S is 5.

Therefore, f(n) of node D = 1. Similarly, node G has h(n) = 0 and the cost to reach node G from node S is 6. Therefore, f(n) of node G = 2.

{(S--> A-->C--->G, 6), (S--> A-->C--->D, 11), (S--> A-->B, 7), (S-->G, 10)}

Iteration 4: Since G is the goal node, the algorithm will return success and terminate giving the final path as, **S--->A--->C--->G** with cost 6.

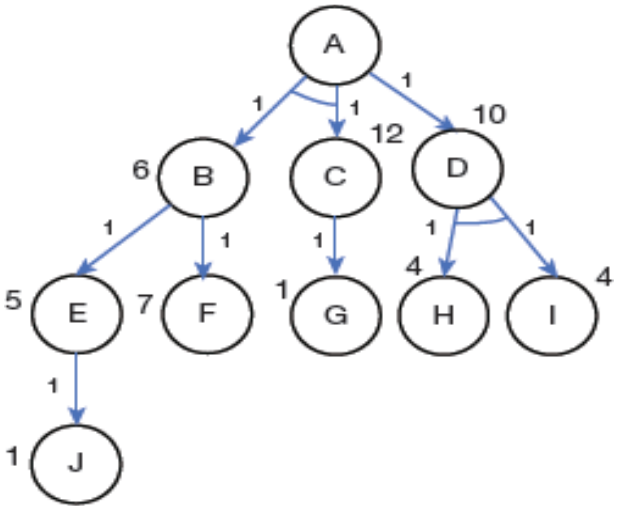
c. AO* Algorithm

- AO algorithm is based on problem decomposition in which a problem is broken down problem into smaller pieces.Therefore, AO* algorithm can be applied only on those algorithms that can be divided into a set of sub problems. Each sub problem can be solved eparately. The final result is obtained by combining all the solutions Usually, AND-OR graphs /trees are used for representing the solution. The decomposition of the problem generates AND arcs.

J. A* Vs AO*

- Both A and AO algorithms are informed search techniques and use heuristic values to solve the problem.
- Although both the algorithms guarantee solution, only A guarantees an optimal solution. By optimal, we mean the shortest path with the lowest cost. The AO algorithm does not always provide an optimal solution. This is because AO does not explore all the solution paths once it finds a solution.

- **Example:** Consider the tree given below and apply the AO* algorithm on it. Remember that



Remember that the AO algorithm always moves towards a lower cost value. The cost function, (F(n)= G (n) + H (n)), where, **H(n) is the**

heuristic or the estimated value of the node and G(n) is the actual cost or edge value.

Step 1: From node A, we can either choose D or B-C together. The cost of both the paths are calculated as,

F(A to D)= 1+10 = 11 and

F(A to B and C) = 1 + 1 + 6 +12 = 20

We will choose node D as the next node to traverse as it has a lower cost than path B-C.

Step 2: From D, we are left with only one choice H-I. Now, the cost of H-I from node D is calculated as,

F(D to H-I) = 1+1+ 4 +4 =10

Correspondingly, the cost of H-I from node A is 11 as,

F(A to D to F-E) = 1+10 =11

If the goal state is found on the path from A to H-I, then the other path remains unexplored.

Let us explore the other path from node A to B-C. We had calculated,

F(A to B and C) = 1 + 1 + 6 +12 = 20

Step 1: From B, there are two nodes can be reached- E and F.

F(B-E)= 5+1 = 6 and F(B-F) = 7 + 1 = 8

Step 2: Since cost from **F(B-F)** is more than that of the path B-E, the algorithm moves to node J.

F(E-J) = 1 +1 = 2, which is less than heuristic value 5. So, the new heuristic value from E is 2.

Step 3: We need to update the cost from E to B, which is now,

F(E-B)= 1+2 =3

Therefore, the new heuristic value of B is 3.

Step 4: We need to update the value of A as A is associated with both B and C.

But to update the heuristic value of A, we must know the value of node C. From C, node G can be reached with a cost of 1. So, the heuristic value of C is now 1 + 1 = 2.

Since cost form C to G = F(C-G) = 2 which is less than its heuristic value, the new heuristic value of C is 2.

Step 5: Now the updated from A- BC that is F(A-BC) = 1+1+2+3 = 7 which is less than F(A - D) = 11.

This shows choosing path A-BC is more cost effective and thus better path A-D.

However, this is possible if the algorithm explores this path as well. But the algorithm will not explore this path. We have just shown this path to demonstrate the weakness of the AO algorithm.

The AO* does not give optimal performance in all the cases.

Advantages

1. It gives optimal solution.
2. It can be used for both OR and AND graph when nodes are traversed in a particular order.

Disadvantages

1. It does not find the optimal solution in every case.
2. Its complexity is more than other algorithms.